

DL Practical 2

IMDB Movie Review Sentiment Analysis using Deep Neural Network

Aim

Classify movie reviews as:

- Positive
- Negative

using Deep Neural Networks and IMDB dataset.

Concept Explanation

This practical is about:

Teaching a Neural Network to understand whether a movie review is Positive or Negative based only on the review text.

This is called:

Binary Classification

Because there are only 2 outputs:

- Positive → 1
- Negative → 0

Output classes:

- 1 → Positive review
- 0 → Negative review

The model learns from text reviews.

What Are We Exactly Doing in Practical 2?

We are building a Deep Neural Network model that learns patterns in movie reviews.

Example:

Positive Review

"Movie was amazing and emotional."

Negative Review

"Worst movie ever."

The model learns:

- Words like "amazing", "excellent" → positive
- Words like "bad", "worst" → negative

After training:

- If we give a new review,
 - The model predicts whether sentiment is positive or negative.
-

Real Flow of the Practical

The practical has 5 major stages:

Stage	What Happens
1	Load IMDB dataset
2	Convert text into numbers
3	Create neural network
4	Train the model
5	Predict sentiment

What is Sentiment Analysis?

Sentiment analysis determines emotional tone from text.

Examples:

- "Amazing movie" → Positive

- "Worst film ever" → Negative
-

Dataset Information

IMDB dataset contains:

- 50,000 movie reviews
 - 25,000 training reviews
 - 25,000 testing reviews
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Step-by-Step Algorithm Explanation

Step 1: Import Libraries

We import:

- keras
- numpy
- tensorflow

Purpose:

- To build and train deep learning model.
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Step 2: Load IMDB Dataset

Dataset contains:

- 50,000 movie reviews
- Labels:
 - 1 → positive
 - 0 → negative

The dataset is already preprocessed in Keras.

Step 3: Reviews Are Stored as Numbers

The text reviews are converted into integer sequences.

Example:

```
[1, 14, 20, 45]
```

Each number represents a word.

Why?

Because neural networks cannot understand raw text directly.

Step 4: One-Hot Encoding / Vectorization

This is one of the MOST IMPORTANT steps.

The review is converted into:

- 0s and 1s vector

Example:

Word	Present?
amazing	1
bad	0
emotional	1

This process is called:

Vectorization

or

One-Hot Encoding

Why We Do This?

Neural Networks only understand:

- numbers

- tensors
- matrices

So we convert text into numeric format.

Step 5: Prepare Labels

Labels are already:

- 1 for positive
- 0 for negative

No extra preprocessing needed.

Step 6: Build Neural Network

The practical creates a Sequential Deep Neural Network.

Architecture:

```
Dense(16,activation='relu')  
Dense(16,activation='relu')  
Dense(1,activation='sigmoid')
```

Explanation of Layers

Hidden Layers

```
Dense(16,activation='relu')
```

Purpose:

- Learn patterns from reviews
 - Detect important word combinations
-

ReLU Activation

Used in hidden layers.

$$\text{ReLU}(x) = \max(0, x)$$

Purpose:

- Adds nonlinearity
 - Helps deep learning model learn complex patterns
-

Output Layer

```
Dense(1,activation='sigmoid')
```

Purpose:

- Gives probability between 0 and 1
-

Sigmoid Function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

If output:

- close to 1 → positive
 - close to 0 → negative
-

Step 7: Compile the Model

The code uses:

```
loss='binary_crossentropy'  
optimizer='rmsprop'  
metrics=['accuracy']
```

Binary Cross entropy

Used because:

- Only two classes exist.

Purpose:

- Measures prediction error.

Smaller loss = better predictions.

Optimizer (RMSProp)

Purpose:

- Updates weights
- Reduces loss
- Helps model learn faster

Similar role to Gradient Descent.

Step 8: Train the Model

The model is trained using:

```
epochs=20  
batch_size=512
```

What Happens During Training?

For every epoch:

1. Model reads reviews
2. Predicts sentiment
3. Calculates error
4. Updates weights
5. Improves predictions

This repeats many times.

Epoch Meaning

1 Epoch =

One complete pass through training dataset.

Batch Size Meaning

512 reviews processed together at once.

Larger batch:

- more stable learning
 - slower updates
-

Step 9: Validation

Part of training data is used for validation.

Purpose:

- Check performance on unseen data
 - Detect overfitting
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Overfitting in This Practical

The graph in the manual shows:

- Training accuracy increasing
- Validation accuracy decreasing

Meaning:

- Model memorized training data
- Poor generalization

This is called: overfitting

Techniques to Reduce Overfitting

1. Dropout
 2. Regularization
 3. Smaller model
 4. Early stopping
-

Final Output

The model predicts:

- Positive review
- Negative review

with around:

- ~89% accuracy

according to the manual

Conclusion

The DNN successfully classifies movie reviews into positive and negative sentiments.

Expected Viva/Oral Questions with Answers

1. What is binary classification?

Classification with only two classes.

Example:

- Positive/Negative
 - Yes/No
-

2. What is sentiment analysis?

Analyzing text to determine emotional tone.

3. Why use sigmoid activation?

Because it gives output between 0 and 1 probability.

4. What is one-hot encoding?

Representing categorical/text data using binary vectors.

5. What is overfitting?

Model performs well on training data but poorly on testing data.

6. What is dropout?

A regularization technique where random neurons are disabled during training.

7. Why use binary cross entropy?

Because output has only two classes.

8. What is ReLU?

Activation function:

$$ReLU(x) = \max(0, x)$$

Used to introduce non-linearity.

9. Difference between ANN and DNN?

DNN has multiple hidden layers, ANN may have fewer.

10. What are epochs and batch size?

- Epoch = One full training cycle

- Batch size = Number of samples processed together
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